

Aadil & Partners Legal

CS5450 Mobile Programming - Challenge 2: Flutter Mobile Portfolio

Group #1: Law Firm Portfolio

App	Aadil & Partners Legal
Tagline	Strategic Legal Counsel. Built on Trust.
GitHub	https://github.com/luqi101/Mobile_Programming_Challenge_2.git
Firebase project	aadil-legal-g1-68e92
Android package	com.aadilpartners.legalportfolio

CS5450 Mobile Programming

Challenge 2: Flutter Mobile Portfolio

Group #1: Law Firm Portfolio

App name: Aadil & Partners Legal

Tagline: Strategic Legal Counsel. Built on Trust. GitHub repository:
https://github.com/luqi101/Mobile_Programming_Challenge_2.git

Group Members

Student ID	Name
1312122	Aadil, Luqman
1339707	Sahil
1307462	ELDELNGATY, ABDELRAHMAN M
1338099	OMAR ALI, AHMED
1311002	ANUJIN, SAINZOLBOO
1297799	Arora, Pranay Rajesh
1340827	Avaiya, Om Jayeshbhai
1332900	Avecillas Segovia, Danilo Nicolas
1296844	Juntao Wen

Project Overview

Aadil & Partners Legal is a professional Flutter/Dart mobile portfolio app for a fictional full-service law firm. The app presents the firm's brand identity, practice areas, fictional attorneys, representative matters, legal resources, FAQ content, and a validated consultation request form.

The project is built as a complete CS5450 Challenge 2 submission. Android emulator support is the primary demonstration target, with Flutter web included as a secondary responsive check. Portfolio content is stored in Cloud Firestore, and the included seed JSON file documents the original fictional data used to populate the database.

Feature Summary

- Firebase-initialized Flutter app with a controlled loading path.
- Professional Material 3 law firm visual design with deep navy, ivory, charcoal, and gold accent colors.
- Responsive mobile layout with bottom navigation for Home, Services, Team, Resources, and Contact.
- Firestore-backed firm profile, practice areas, attorneys, representative matters, testimonials, legal resources, and FAQs.
- Practice area cards and detail screens for eight law firm service categories.
- Fictional attorney profiles with credentials, experience, specialization, and detail screens.
- Original legal resources and expandable FAQ content.
- Consultation request form with full name, email, optional phone, preferred practice area, message, and consent acknowledgement.
- Valid contact submissions are written to the Firestore `contact_requests` collection with a server timestamp and `source: "mobile_app"`.
- Debug-only seed utility and a CLI seeding script for reproducible Firestore setup.
- Local seed JSON fallback exists only to prevent a broken demo state if Firestore is temporarily unavailable.

Firebase Configuration

This project uses:

- `firebase_core` for Firebase initialization.
- `cloud_firestore` for portfolio data reads and consultation request writes.

Configured Firebase project:

- Firebase project ID: `aadil-legal-g1-68e92`
- Firebase display name: `aadil-legal-g1`
- Android package/application ID: `com.aadilpartners.legalportfolio`
- Firestore database: `(default)`
- Firestore region: `northamerica-northeast1`

The Flutter app initializes Firebase in `lib/main.dart` with:

```
Firebase.initializeApp(  
  options: DefaultFirebaseOptions.currentPlatform,  
)
```

The real FlutterFire configuration is generated locally and intentionally ignored by Git:

```
lib/firebase_options.dart  
android/app/google-services.json
```

Safe examples are committed for documentation:

```
lib/firebase_options.example.dart
```

android/app/google-services.example.json

No Firebase service account keys, private keys, or sensitive credentials are included in this repository.

Firestore Data Model

Collection	Purpose
firm_profile/default	Firm name, tagline, overview, mission, values, stats, contact information, and disclaimer
practice_areas	Law firm service areas and detail content
attorneys	Fictional lawyer profiles
representative_matters	Anonymized fictional representative matters
testimonials	Fictional testimonials using client initials only
resources	Original legal insight/resource articles
faqs	Frequently asked questions and answers
contact_requests	Validated consultation form submissions

The portfolio collections are publicly readable for the app demo. Public writes to portfolio collections are denied. The `contact_requests` collection allows create-only writes with validation and denies public read, update, and delete access.

Firestore configuration files:

```
firebase.json
firestore.rules
firestore.indexes.json
```

Deploy rules and indexes with:

```
firebase deploy --only firestore:rules,firestore:indexes --project aadil-legal-g1-68e92
```

Firestore Seeding

Seed data is stored in:

```
assets/data/portfolio_seed.json
```

The project includes a CLI seeding utility:

```
tool/seed_firestore.dart
```

Run the seeder after Firebase configuration and Firestore database creation:

```
dart run tool/seed_firestore.dart
```

The seeding workflow writes deterministic documents for:

- `firm_profile`
- `practice_areas`
- `attorneys`
- `representative_matters`

- testimonials
- resources
- faqs

The seeder does not create real contact requests. Contact requests are created only through validated form submissions or explicit validation testing.

Additional notes are available in:

`docs/firebase_seed_instructions.md`

Setup Instructions

Install prerequisites:

- Flutter stable SDK
- Android Studio with Android SDK and at least one Android emulator
- Firebase CLI
- FlutterFire CLI

Verify Flutter:

```
flutter --version
flutter doctor
```

Install dependencies:

```
flutter pub get
```

If Firebase needs to be reconfigured on another machine:

```
firebase login
dart pub global activate flutterfire_cli
flutterfire configure \
  --yes \
  --project=aadil-legal-g1-68e92 \
  --platforms=android,web \
  --android-package-name=com.aadilpartners.legalportfolio \
  --out=lib/firebase_options.dart
```

This command recreates both `lib/firebase_options.dart` and `android/app/google-services.json`. These generated files are required to run the app locally, but they are not committed to the public GitHub repository because they contain Firebase client API keys.

Set the Firebase project:

```
firebase use aadil-legal-g1-68e92
```

Deploy Firestore rules:

```
firebase deploy --only firestore:rules,firestore:indexes --project aadil-legal-g1-68e92
```

Run Instructions

Run on Android emulator:

```
flutter run
```

Build an Android debug APK:

```
flutter build apk --debug
```

Optional Chrome responsive check:

```
flutter run -d chrome
```

Android emulator execution is the primary demonstration path for this submission. Chrome/web is included only as an additional platform check.

Project Structure

```
.
|-- AGENTS.md
|-- MC_Challenge2_2026NN.pdf
|-- README.md
|-- README.pdf
|-- analysis_options.yaml
|-- android/
|   |-- app/
|   |   |-- google-services.example.json
|-- assets/
|   |-- data/
|   |   |-- portfolio_seed.json
|-- docs/
|   |-- firebase_seed_instructions.md
|   |-- screenshots/
|   |   |-- 01_home.png
|   |   |-- 02_practice_areas.png
|   |   |-- 03_practice_areas_more.png
|   |   |-- 04_practice_area_detail.png
|   |   |-- 05_team.png
|   |   |-- 06_attorney_detail.png
|   |   |-- 07_resources.png
|   |   |-- 08_resource_detail.png
|   |   |-- 09_contact_form.png
|   |   |-- 10_faq_collapsed.png
|   |   |-- 11_faq_expanded.png
|   |-- README.md
|-- firebase.json
|-- firestore.indexes.json
|-- firestore.rules
|-- ios/
|-- lib/
|   |-- app.dart
|   |-- firebase_options.example.dart
|   |-- main.dart
|   |-- core/
|   |-- data/
|   |-- features/
|   |-- shared/
|-- pubspec.lock
|-- pubspec.yaml
|-- screenshots/
|-- test/
|-- tool/
|   |-- build_readme_pdf.py
|   |-- seed_firestore.dart
|-- web/
```

Screenshots

The following screenshots were captured from the Android app and are included in docs/screenshots/.

Validation Results

Final validation completed successfully:

- flutter pub get passed.
- dart format . passed.
- flutter analyze passed with no issues found.
- flutter test passed.
- flutter build apk --debug passed and produced build/app/outputs/flutter-apk/app-debug.apk.
- Android emulator run was tested successfully by the project owner after Firebase configuration and seeding.
- Firestore portfolio seed verification showed documents in firm_profile, practice_areas, attorneys, representative_matters, testimonials, resources, and faqs.
- Contact request write behavior was verified against the contact_requests collection.

Demo Flow

Recommended Zoom demonstration:

- Open the app on an Android emulator.
- Show the Home screen and confirm the app presents Aadil & Partners Legal.
- Navigate to Services and open a practice area detail screen.
- Navigate to Team and open an attorney profile.
- Navigate to Resources and open a resource article.
- Open the FAQ screen from Resources.
- Navigate to Contact and submit a valid consultation request.
- Confirm the request appears in the Firestore contact_requests collection.
- Mention that all content, branding, biographies, testimonials, matters, and articles are fictional and original.

ZIP Packaging Instructions

Include:

- lib/
- android/
- ios/
- web/

- assets/
- docs/screenshots/
- screenshots/
- pubspec.yaml
- pubspec.lock
- README.md
- README.pdf
- firebase.json
- firestore.rules
- firestore.indexes.json
- all Dart source files and tests

Do not include:

- build/
- .dart_tool/
- .gradle/
- backup folders
- private keys
- Firebase service account JSON
- local machine caches

Academic Originality Statement

All law firm branding, attorney biographies, testimonials, representative matters, resource text, and UI composition are fictional and created for this project. The app does not copy real law firm branding, real attorney profiles, real client testimonials, or unlicensed images.

Legal Disclaimer

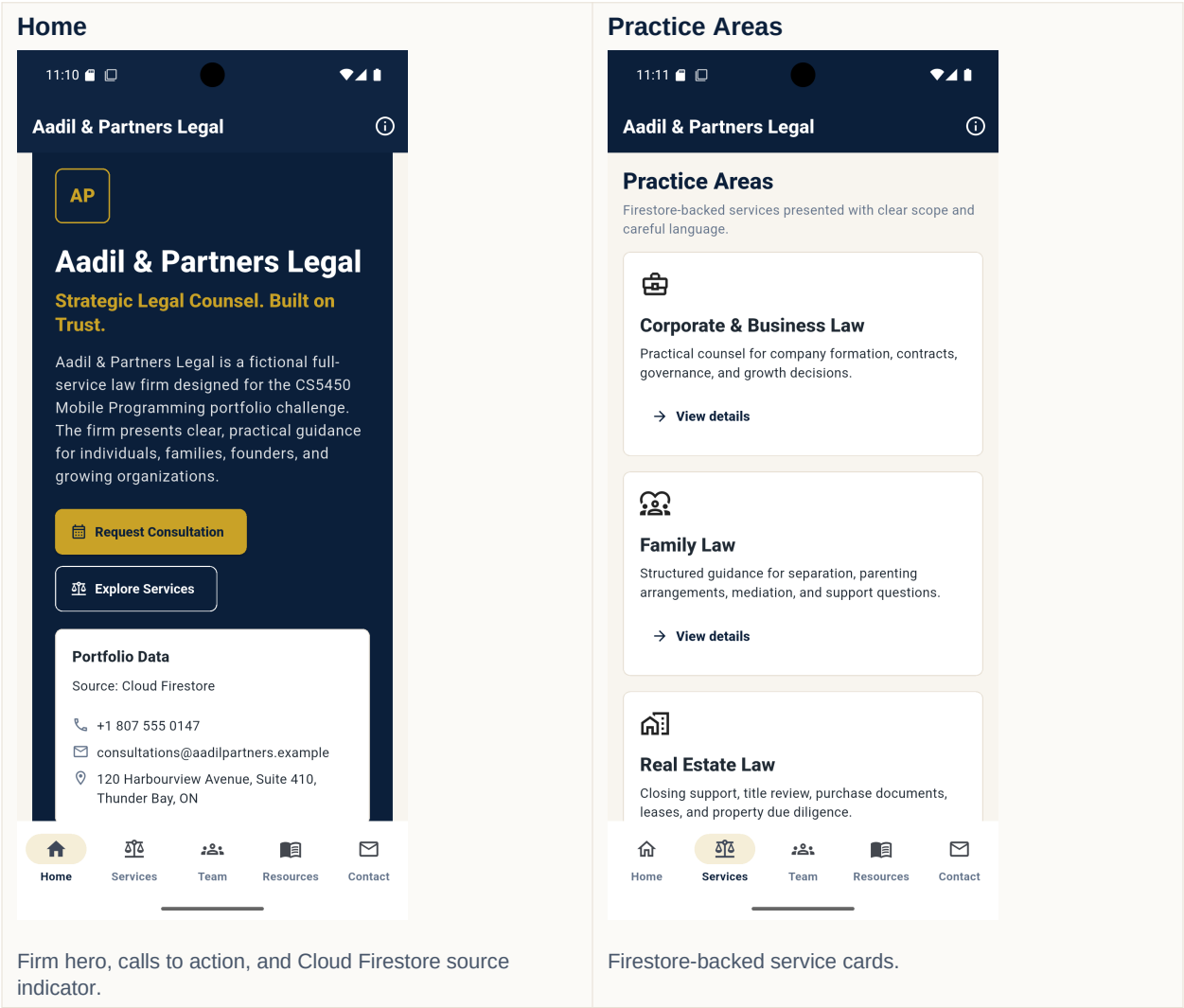
This app is an educational student portfolio project. It does not provide legal advice. Submitting the consultation request form does not create a lawyer-client relationship. Representative matters are fictional examples and do not guarantee any result.

Known Limitations

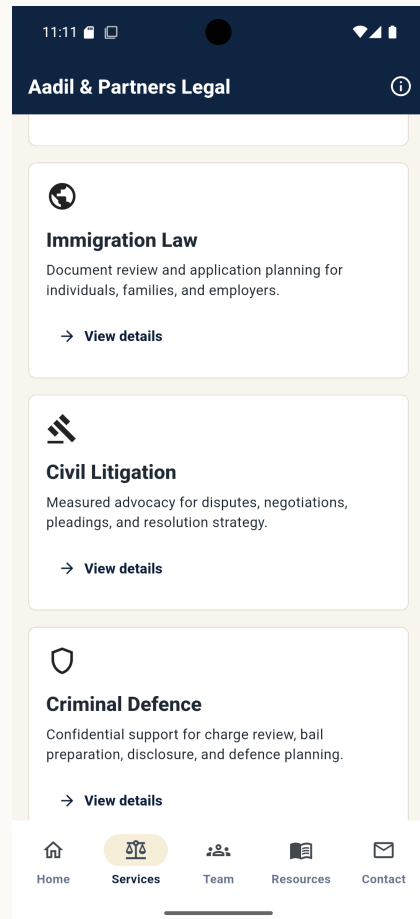
No build-blocking limitations are known for the Android emulator demonstration. The app is configured for Android and web through FlutterFire; Android is the primary graded demo target.

Screenshot Appendix

The following Android screenshots are bundled with the project and referenced by the README.

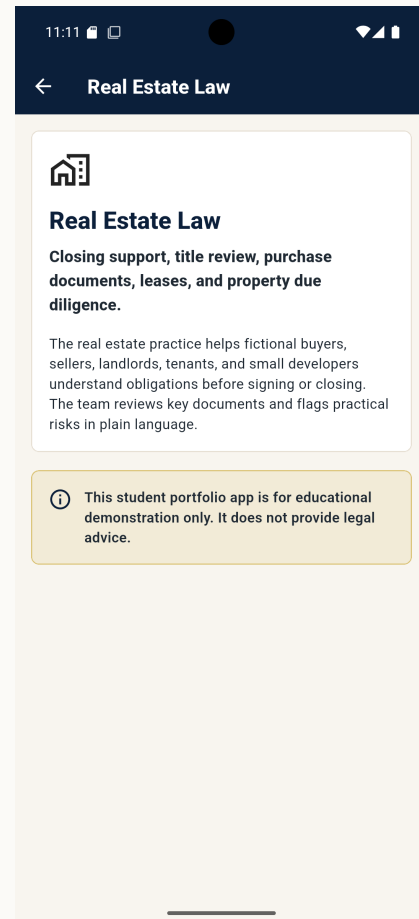


Services List



Additional practice area cards.

Service Detail



Practice area detail with educational disclaimer.

Team

11:11

Aadil & Partners Legal

Attorneys

Fictional lawyers with original bios and portfolio-safe credentials.

SA

Samira Aadil
Managing Partner
Corporate governance and civil disputes
14 yrs J.D., called to the fictional Ontario bar

DH

Daniel Hart
Partner
Family law and estates
11 yrs J.D., mediator training certificate

MC

Mira Chen

Home

Services

Team

Resources

Contact

Fictional attorney portfolio cards.

Attorney Detail

11:12

← Daniel Hart

DH

Daniel Hart
Partner

Family law and estates

Daniel works with fictional families on practical agreements, planning documents, and respectful dispute resolution. He emphasizes preparation, transparency, and careful communication.

Credentials

J.D., mediator training certificate

Education

North Shore Faculty of Law, J.D.

Experience

11 years

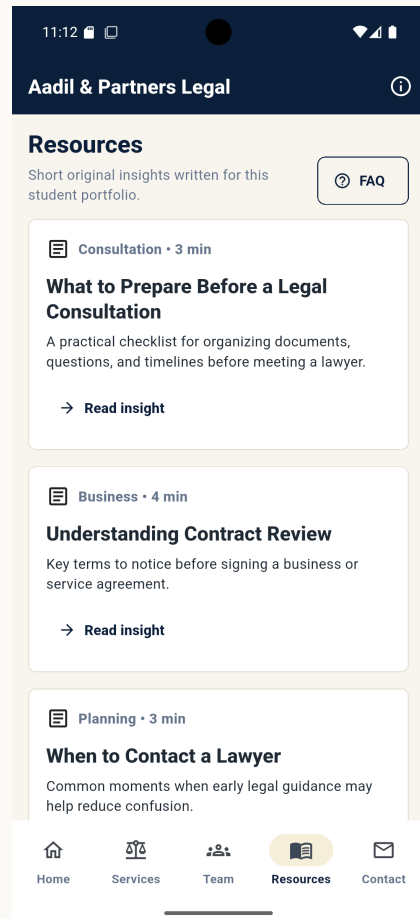
Email

daniel.hart@aadilpartners.example

Open email app

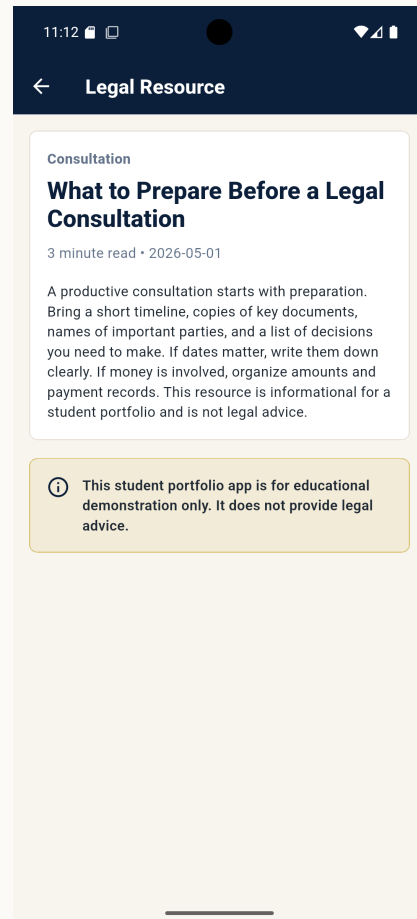
Attorney profile detail and contact action.

Resources



Original legal insight cards.

Resource Detail



Legal resource article view.

Contact Form

11:12

Aadil & Partners Legal

Request a Consultation

Validated intake form. Successful submissions are written to Firestore.

 Full name

 Email

 Phone optional

 Preferred practice area

Message

☐ Submitting this form does not create a lawyer-client relationship. A formal engagement requires conflict checks and written acceptance by the firm.

Submit request

Home

Services

Team

Resources

Contact

Validated consultation request form.

FAQ

11:12

FAQ

Frequently Asked Questions

Practical intake questions for a fictional law firm portfolio.

Does submitting the consultation form create a lawyer-client relationship?

Intake

What documents should I bring to a consultation?

Consultation

How are legal fees discussed?

Fees

Is the consultation confidential?

Privacy

How long does a matter take?

Process

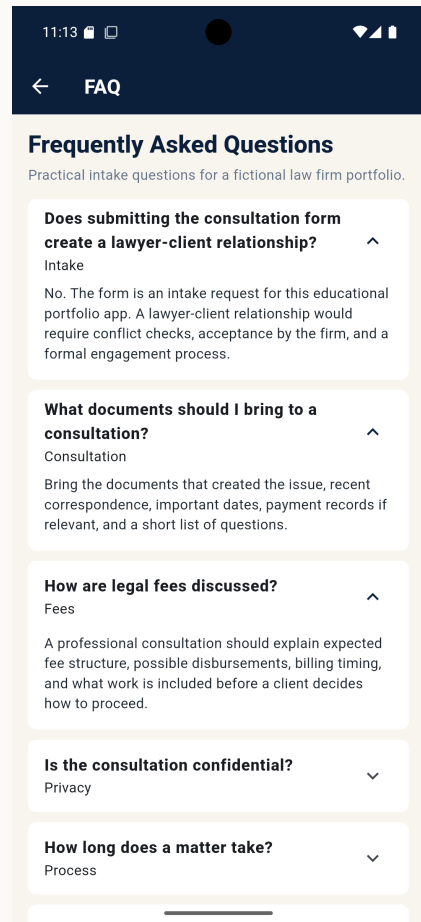
Can consultations happen remotely?

Access

 This student portfolio app is for educational demonstration only. It does not provide legal advice.

Firestore-backed FAQ list.

Expanded FAQ



Expanded FAQ answers.